

ORIGINAL ARTICLE

Effectiveness of zinc supplementation in laryngopharyngeal reflux assessed by reflux symptom index and reflux finding score

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ABSTRACT

Background and aim: Standard therapy for Laryngopharyngeal Reflux (LPR) with Proton Pump Inhibitors (PPIs) frequently yields suboptimal responses, particularly in reversing laryngeal mucosal injury. This study aimed to evaluate the clinical effectiveness of adjunctive zinc supplementation in patients receiving PPI therapy.

Methods: An experimental pre–post control group study was conducted involving 22 patients diagnosed with LPR (Reflux Symptom Index [RSI] ≥ 13 and Reflux Finding Score [RFS] ≥ 7). Participants were allocated into two groups. The treatment group received Omeprazole 20 mg twice daily plus Zinc Sulfate 20 mg once daily, while the control group received Omeprazole 20 mg twice daily plus placebo for 12 weeks. Treatment effectiveness was assessed using changes in RSI and RFS scores. Statistical analyses were performed using the Mann–Whitney U test and Wilcoxon signed-rank test, with $p < 0.05$ considered statistically significant.

Results: Both groups demonstrated post-treatment improvement. However, the zinc group exhibited significantly greater reductions in clinical scores. The mean reduction (Δ) in RSI score in the zinc group was 7.46 ± 1.7 compared with 2.91 ± 2.9 in the control group ($p = 0.003$). The mean reduction in RFS score was 3.72 ± 1.7 in the zinc group versus 0.73 ± 1.5 in controls ($p = 0.001$). Laryngeal edema resolution was more consistently observed in the zinc group.



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Conclusions: Combination of PPI and zinc supplementation is superior to PPI monotherapy in improving both subjective symptoms and objective laryngoscopic findings in LPR patients. Zinc appears to be an effective adjuvant therapy that accelerates mucosal inflammatory resolution. (www.actabiomedica.it)

Keywords: laryngopharyngeal reflux, zinc, proton pump inhibitor, reflux symptom index, reflux finding score

Introduction

Laryngopharyngeal Reflux (LPR) is defined as the retrograde flow of gastric contents into the laryngopharynx, resulting in exposure of the upper aerodigestive tract mucosa to acidic and enzymatic components. Unlike the esophagus, the laryngopharyngeal mucosa demonstrates limited intrinsic protective mechanisms against acid and pepsin exposure, rendering it particularly susceptible to chronic inflammation, laryngitis, and potentially premalignant lesions (1,2).

Epidemiological trends indicate a growing prevalence of LPR, particularly in association with modern lifestyle changes and dietary habits (3). In Asian populations, including Indonesia, hospital-based data suggest that approximately 20–30% of patients presenting with laryngeal complaints are diagnosed with LPR. The disease substantially impairs quality of life, affecting both physical comfort and psychosocial well-being (4).

Empirical therapy with Proton Pump Inhibitors (PPIs) remains the cornerstone of LPR management (5). However, increasing evidence questions their overall effectiveness, particularly in improving objective laryngoscopic findings. While PPIs effectively suppress gastric acid secretion, they do not directly inactivate pepsin or repair established mucosal injury. Moreover, long-term high-dose PPI therapy has been associated with adverse effects, including micronutrient malabsorption and increased infection risk (3,6).

These therapeutic limitations underscore the need for adjunctive strategies targeting mucosal defense and repair. Zinc, an essential trace element, plays a critical role in cellular proliferation, DNA synthesis, and immune modulation. Zinc serves as a cofactor for carbonic anhydrase, a metalloenzyme involved in mucosal buffering mechanisms within the laryngopharynx (3,7).

Additionally, zinc exhibits cytoprotective and anti-inflammatory properties that may facilitate epithelial regeneration following reflux-induced injury (8,9).

Although theoretical and experimental evidence supports zinc's role in mucosal healing, clinical data evaluating its effectiveness in LPR—particularly in Indonesian populations—remain limited. This study aimed to assess the clinical impact of zinc supplementation as an adjunct to PPI therapy in LPR patients using validated subjective (Reflux Symptom Index/RSI) and objective (Reflux Finding Score/RFS) assessment tools.

Materials and methods

Study design and setting

This quasi-experimental study employed a pre-post test control group design utilizing systematic allocation. The research was conducted at the Department of Otolaryngology–Head and Neck Surgery, Dr. Wahidin Sudirohusodo Hospital and Hasanuddin University Teaching Hospital, Makassar, Indonesia, from April to December 2025. Ethical approval was obtained from the Institutional Review Board of the Faculty of Medicine, Hasanuddin University (No. 410/UN4.6.4.5.31/PP36/2025). Written informed consent was obtained from all participants.

Participants

Eligible participants were adults aged 18–60 years diagnosed with LPR based on RSI ≥ 13 and RFS ≥ 7 . Exclusion criteria included a history of head and neck malignancy, structural laryngopharyngeal or

esophageal abnormalities, prior radiotherapy or chemotherapy, and known zinc allergy.

Thirty-eight patients were initially recruited through consecutive sampling. Sixteen patients were excluded due to non-adherence or loss to follow-up, resulting in 22 patients completing the study (11 per group).

Intervention

Participants were allocated systematically according to their order of presentation using an odd-even numbering method. The treatment group received Omeprazole 20 mg twice daily in combination with Zinc Sulfate Monohydrate 20 mg once daily, whereas the control group received Omeprazole 20 mg twice daily plus a placebo. The therapeutic regimen was administered for 12 weeks. Medication adherence was monitored daily through telephone calls or text messaging to ensure compliance throughout the study period.

Outcome measures

Clinical evaluation was conducted at baseline and after 12 weeks of therapy using two validated instruments. The RSI was used as a subjective assessment tool (10). RSI is a validated 9-item self-administered questionnaire designed to evaluate the severity of LPR-related symptoms, with a total score ≥ 13 considered abnormal (10,11). Objective assessment was performed using the RFS, which is based on fiberoptic nasopharyngolaryngoscopy and evaluates eight laryngeal parameters, with a total score ≥ 7 indicating abnormal findings consistent with LPR. All RFS assessments were performed by a single, experienced otolaryngologist who was blinded to the participants' treatment group allocations. This approach was strictly implemented to minimize inter-observer variability, ensure consistency in scoring, and mitigate the risk of detection bias.

Statistical analysis

Data were analyzed using SPSS version 25. Continuous variables were expressed as mean $\pm$ standard deviation. Between-group comparisons were conducted using the Independent t-test for normally distributed data or the Mann-Whitney U test for non-normally distributed data. Within-group comparisons

of pre- and post-treatment scores were analyzed using the Wilcoxon signed-rank test. A p-value of less than 0.05 was considered statistically significant.

Results

A total of 38 participants were initially recruited; however, final analysis was performed on 22 participants who completed the entire study protocol (11 in the treatment group and 11 in the control group).

Baseline characteristics are presented in Table 1. The majority of participants were female (81.8% in the treatment group and 100% in the control group). The mean age in the treatment group was 44.6 ± 10.4 years, compared with 41.5 ± 11.3 years in the control group. Statistical analysis revealed no significant differences between groups in terms of age or sex distribution ($p > 0.05$), indicating that both groups were demographically comparable at baseline.

Reflux Symptom Index (RSI)

Evaluation of subjective symptom severity using the RSI demonstrated significant improvement in both groups after 12 weeks of therapy (Table 2). In the treatment group receiving PPI plus zinc, the mean RSI score decreased significantly from 17.55 ± 3.98 at baseline to 10.09 ± 1.97 at 3 months ($p = 0.001$) (Figure 1). Similarly, the control group exhibited a significant reduction from 17.18 ± 4.49 to 14.27 ± 2.00 ($p = 0.005$).

Despite improvements in both groups, between-group comparison of the mean reduction (Δ RSI) demonstrated a markedly greater decrease in the treatment group (7.46 ± 1.7) compared with the control group (2.91 ± 2.9), with the difference reaching statistical significance ($p = 0.003$). These findings indicate that zinc supplementation provided superior reduction in subjective LPR symptoms compared with PPI monotherapy.

Reflux Finding Score (RFS)

Objective assessment using nasopharyngolaryngoscopy to determine the RFS showed a pattern consistent with subjective improvement. In the treatment group, the mean RFS score decreased significantly from 9.36 ± 2.2 to 5.64 ± 1.4 ($p = 0.003$) (Figure 2).

Table 1. Characteristics of participants.

Variable	Treatment Group (n = 11)	Control Group (n = 11)	p-value
Age (years)			
Mean $\pm$ SD	44.6 $\pm$ 10.4	41.5 $\pm$ 11.3	0.493
Sex, n (%)			
Male	2 (18.2%)	0 (0%)	0.476
Female	9 (81.8%)	11 (100%)	

Note: Independent *t*-test for age; Fisher's exact test for sex.

Table 2. Pre- and post-therapy changes in RSI and RFS scores.

Variable	Pre-Therapy (Mean $\pm$ SD)	3-Month Post-Therapy (Mean $\pm$ SD)	p-value (Within Group)	Mean Reduction (Δ)	p-value (Between Groups)
RSI Score					
Treatment	17.55 $\pm$ 3.98	10.09 $\pm$ 1.97	0.001	7.46 $\pm$ 1.7	0.003
Control	17.18 $\pm$ 4.49	14.27 $\pm$ 2.00	0.005	2.91 $\pm$ 2.9	
RFS Score					
Treatment	9.36 $\pm$ 2.2	5.64 $\pm$ 1.4	0.003	3.72 $\pm$ 1.7	0.001
Control	10.18 $\pm$ 1.7	9.45 $\pm$ 1.4	0.007	0.73 $\pm$ 1.5	

Note: Wilcoxon signed-rank test for within-group comparisons; Mann-Whitney *U* test for between-group comparisons.

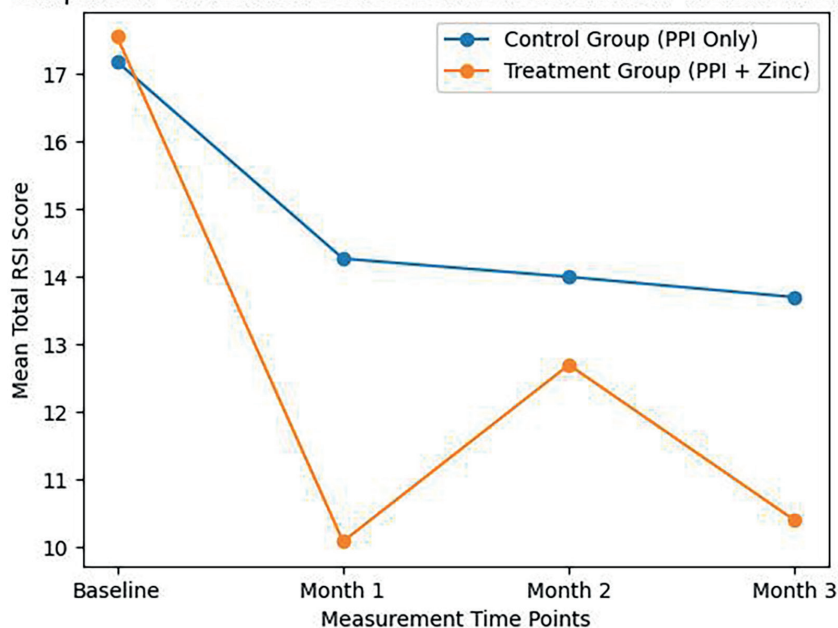
Comparison of Mean RSI Scores Between Control and Treatment Groups

Figure 1. Comparison of mean Reflux Symptom Index (RSI) scores between the treatment and control groups at baseline, 1 month, 2 months, and 3 months. Both groups demonstrated a reduction in RSI scores over time; however, the treatment group (PPI plus zinc) showed a more pronounced and sustained decrease compared with the control group receiving PPI alone, particularly at 3 months of therapy.

In the control group, a statistically significant but modest reduction was observed, from 10.18 ± 1.7 to 9.45 ± 1.4 ($p = 0.007$) (Table 2).

Between-group comparison revealed a substantially greater mean reduction (Δ RFS) in the treatment group (3.72 ± 1.7) compared with the control group (0.73 ± 1.5). The Mann–Whitney U test confirmed that this difference in laryngoscopic improvement was highly significant ($p = 0.001$).

Specifically, morphological improvement in the zinc group was most evident in parameters related to vocal fold edema and subglottic edema, whereas changes in the control group were minimal.

Discussion

The present study strengthens the growing body of clinical evidence indicating that adjunctive zinc

supplementation combined with standard PPI therapy provides superior clinical improvement compared with PPI monotherapy in patients with LPR. The most salient finding of this investigation is the magnitude of reduction (delta change) in both RSI) and RFS observed in the zinc combination group, which was consistently and significantly greater than that in the control group. Although both groups demonstrated improvement following three months of therapy, the control group receiving PPI alone exhibited only minimal improvement in objective laryngeal signs (RFS reduction <1 point).

These findings align with the study by Guo et al. (5) and Lechien et al. (4), which reported that PPI therapy often fails to significantly reverse laryngeal mucosal injury compared with placebo, despite its efficacy in alleviating heartburn symptoms. The relative limitation of standard therapy may be explained by the multifactorial pathophysiology of LPR, which is

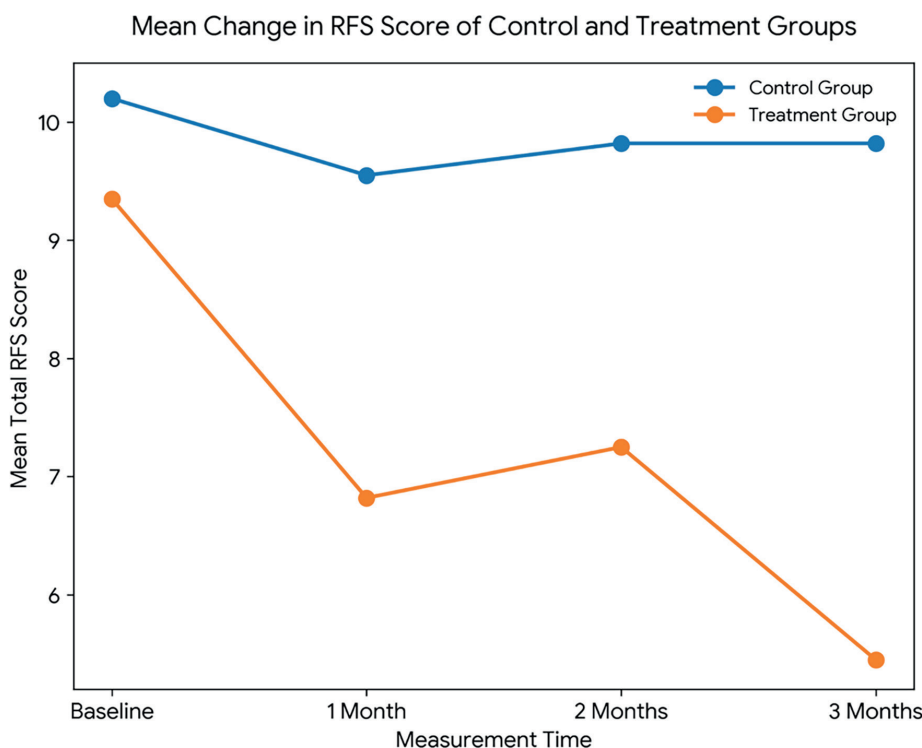


Figure 2. Comparison of mean Reflux Finding Score (RFS) scores between the treatment and control groups at baseline, 1 month, 2 months, and 3 months. While the control group showed minimal changes in RFS scores over time, the treatment group (PPI plus zinc) demonstrated a substantial and progressive reduction. The decrease in RFS scores was significantly more pronounced in the treatment group compared with the control group receiving PPI alone, with the greatest divergence observed at 3 months of therapy.

not solely dependent on acid exposure. Pepsin, a proteolytic gastric enzyme, remains stable and potentially active at pH levels up to 6.5 and may undergo intracellular reactivation within laryngeal mucosal compartments through pepsin endocytosis (12). PPIs elevate gastric pH but do not irreversibly inactivate pepsin nor restore epithelial barrier integrity once disrupted (13). Consequently, there is a compelling need for therapeutic agents that act at the level of mucosal defense—an effect that, in this study, appears to be fulfilled by zinc supplementation.

The clinical effectiveness of zinc observed in this study can be explained by its pleiotropic mechanisms of action. First, zinc plays a critical role in restoring laryngeal buffering capacity. Physiologically, the laryngopharyngeal mucosa has limited intrinsic defense against refluxate due to lower bicarbonate production compared with the esophagus. Zinc functions as an essential cofactor for the enzyme carbonic anhydrase (CA), particularly the CA-III isoenzyme, which is highly expressed in laryngeal tissue (3,7). This enzyme catalyzes the hydration of carbon dioxide to bicarbonate, thereby neutralizing acidity in both extracellular and intracellular compartments (14). Under chronic inflammatory conditions, local zinc depletion may impair CA activity; zinc supplementation may therefore help reestablish local pH homeostasis.

Second, the significant reduction in RFS—particularly the resolution of vocal fold edema and subglottic edema—observed in the treatment group is closely associated with zinc's anti-inflammatory properties. Zinc has been shown to inhibit the Nuclear Factor-kappa B (NF- κ B) signaling pathway, a central regulator of pro-inflammatory cytokine transcription, including TNF- α , IL-1 β , and IL-6 (15–17). By attenuating this cytokine cascade, zinc reduces vasodilation and vascular permeability, which clinically manifest as laryngeal edema. Furthermore, zinc contributes to mast cell membrane stabilization, thereby inhibiting degranulation and histamine release, which may accelerate the resolution of chronic inflammatory signs reflected in the RFS.

Third, zinc exerts a protective effect on tight junction integrity. Mucosal injury in LPR is often initiated by disruption of tight junction proteins (e.g., E-cadherin and occludin) secondary to pepsin exposure, resulting

in increased mucosal permeability (“leaky mucosa”) (18,19). Zinc has been demonstrated to modulate tight junction protein expression and reinforce epithelial barrier function, thereby preventing further penetration of irritants into the submucosal layer (18–20).

In the context of subjective symptom improvement (RSI), the zinc group exhibited a marked reduction in specific complaints, particularly globus sensation and the frequent urge for throat clearing. These findings are consistent with the study by Pramana et al., which reported significant clinical improvement following zinc supplementation (21). This clinical enhancement is likely a secondary effect of accelerated mucosal re-epithelialization. Zinc serves as a critical structural and catalytic component for DNA and RNA polymerase enzymes, which are essential for cell proliferation and protein synthesis during wound healing and tissue regeneration (22,23).

The ability of zinc to expedite the closure of microscopic epithelial defects in the larynx prevents recurrent exposure of laryngeal sensory nerve endings (irritant receptors). This process subsequently diminishes laryngeal hypersensitivity, which frequently manifests as neurogenic chronic cough or persistent throat clearing. Furthermore, the 3-month (12-week) study duration was a pivotal factor; in contrast to short-term studies (4 weeks), this extended period provided adequate time for zinc to facilitate a complete epithelial regeneration cycle, resulting in more pronounced clinical outcomes compared to the control group.

The interpretation of these results must be considered within the context of several limitations. First, the study employed a quasi-experimental design using systematic (odd-even) allocation rather than true randomization, which inherently introduces a potential risk of selection bias. Second, the relatively small final sample size and the exceptionally high attrition rate (~42%) significantly limit the generalizability of the findings and introduce a potential risk of attrition bias. The high dropout rate was primarily attributed to patients' reluctance to undergo the repeat nasopharyngolaryngoscopy at the 12-week follow-up, particularly among those who had already experienced early symptomatic relief. Because complete paired data were required for the objective RFS evaluation, a strictly per-protocol analysis was conducted rather than an

intention-to-treat analysis, which may overestimate the treatment effect. Third, the diagnosis of LPR was established based on clinical criteria (RSI and RFS) without confirmation via Multichannel Intraluminal Impedance-pH monitoring (MII-pH)—the current gold standard—due to facility constraints at the research center. Fourth, subjects' serum zinc levels were not measured before or after therapy. Consequently, we could not determine whether the subjects presented with baseline systemic zinc deficiency or if the observed therapeutic effects were purely local pharmacological actions. Finally, the inherent subjectivity in RFS assessment remains a potential source of bias, although this was minimized by utilizing a single, experienced blinded assessor.

Despite these limitations, this study offers a critical clinical perspective: the management of LPR should not rely solely on gastric acid suppression. Approaches targeting mucosal defense and repair (mucosal protective agents) are essential to disrupt the cycle of chronic inflammation. Zinc sulfate, as an affordable and safe supplement with a minimal side-effect profile, holds significant potential as a routine adjuvant therapy. Its use is highly recommended, particularly for LPR patient phenotypes exhibiting a partial response to PPIs or those with predominant mucosal irritation, such as persistent edema and erythema.

Conclusion

The combination of Zinc Sulfate and PPI therapy proved significantly more effective in improving clinical symptoms (RSI) and laryngoscopic findings (RFS) compared to PPI monotherapy. These results confirm that the management of LPR requires mucosal cytoprotective agents rather than simple acid suppression alone. Consequently, zinc is recommended as a routine adjuvant therapy to accelerate the resolution of inflammation. Future studies should employ a double-blind, RCT design with larger sample sizes and incorporate MII-pH monitoring as the diagnostic gold standard. Additionally, the analysis of inflammatory biomarkers (such as salivary pepsin and cytokines) and serum zinc levels is necessary to objectively validate the underlying biomolecular mechanisms.

Ethic approval: All research designs were reviewed and approved by the Health Research Ethics Committee of Dr Wahidin Sudirohusodo Hospital, Faculty of Medicine, Hasanuddin University (No. 410/UN4.6.4.5.31/PP36/2025) on June 17, 2025.

Conflict of interest: Each author declares that he or she has no commercial associations that might pose a conflict of interest in connection with the submitted article.

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Data availability statement: The data presented in this study are available on request from the corresponding author.

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